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Magnetoviscosity of dilute magnetic fluids in oscillating and rotating magnetic fields

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The magnetic field dependent viscosity (magnetoviscosity) of dilute suspensions of spherical magnetic particles suspended in a Newtonian fluid subjected to both magnetic and shear flow fields was studied through Brownian dynamics simulations. Results are presented for the response of these suspensions to oscillating magnetic fields and magnetic fields that are corotating or counter-rotating with respect to the fluid vorticity. A decrease in negative values in the intrinsic magnetoviscosity is observed for oscillating and corotating magnetic fields, whereas an increase is observed for counter-rotating magnetic fields. The frequency corresponding to zero viscosity and the minimum value in the negative viscosity are lower for corotating magnetic fields than for oscillating magnetic fields. In the negative magnetoviscosity regions the particles in a corotating magnetic field rotate faster than in an oscillating magnetic field. It is estimated that the flow due to corotating particles could be strong enough to obtain a negative effective viscosity in dilute suspension. Moreover, it is shown that the commonly accepted constitutive equation for the antisymmetric stress describes well the intrinsic magnetoviscosity of the suspension. © *2010 American Institute of Physics*.
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I. INTRODUCTION

Colloidal dispersions of magnetic nanoparticles in a liquid carrier, so-called ferrofluids or magnetic fluids,[1] show interesting rheological and magnetic properties when they are influenced by an external magnetic field.[2] In a simple shear flow field it is known that if an external magnetic field is applied, the particles will rotate relative to the local vorticity of the fluid, resulting in a change in the viscosity of the suspension. This change in viscosity is called magnetoviscosity[3] or rotational viscosity[2,4] and is in addition to the increase in viscosity for force- and torque-free suspended particles, first estimated by Einstein[5,6] for a suspension of rigid spherical particles.

The increase in the viscosity of a magnetic fluid under the influence of a constant magnetic field was first observed experimentally by McTague[3] and Rosensweig *et al.*[7] In the presence of a stationary magnetic field the rotation of the magnetic particles is hindered in a shear flow; therefore the fluid is forced to flow around the particles which leads to a greater rate of mechanical energy dissipation that results in a larger apparent viscosity.[8] Shliomis[4] developed a theoretical expression for the magnetoviscosity in stationary magnetic fields based on macroscopic equations of motion and magnetization. Later, modified magnetization equations derived from the Fokker–Planck equation and irreversible thermodynamics were used to improve the model for the magnetoviscosity.[9] On the other hand, the rheological properties of suspensions of spherical[10,11] and nonspherical[12–15] particles have been studied numerically from a microscopic point of view.

The effect of an alternating magnetic field on the rheological properties of magnetic fluids has also been considered. In this case, under certain conditions the magnetic field reinforces the rotational motion of the particles and part of the energy of the magnetic field is transferred to the fluid as kinetic energy, accelerating the ferrofluid flow. This effect is seen as if the viscosity of the suspension decreased.[16,17] Shliomis and Morozov[18] showed theoretically that if the magnetic field oscillates with a high enough frequency, $\Omega\tau_B > 1$, where Ω is the oscillating field frequency and $\tau_B = 3\eta_0 V_h/(k_B T)$ is the Brownian relaxation time, the magnetoviscosity of a dilute suspension of spherical particles becomes negative. The expression derived by Shliomis and Morozov for the magnetoviscosity η^m in an oscillating magnetic field of small amplitude is given by

$$\eta^m = \frac{1}{4}\eta_0\phi\alpha^2\frac{1-\Omega^2\tau_B^2}{(1+\Omega^2\tau_B^2)^2}, \tag{1}$$

where η_0 is the viscosity of the carrier fluid, ϕ is the volume fraction of particles, and $\alpha = \mu_0 mH/k_B T$ is the Langevin parameter, with μ_0 being the magnetic permeability of free space, m the particle's magnetic dipole moment, H the magnetic field, k_B Boltzmann's constant, and T the absolute temperature.

The so-called negative viscosity effect, a decrease in ferrofluid viscosity in an oscillating magnetic field, was first confirmed experimentally by Bacri *et al.*[19] for a water-based magnetic fluid composed of cobalt-ferrite ($CoFe_2O_4$) particles in the Poiseuille flow and subjected to an axial oscil-

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lating magnetic field. They observed a decrease in the magnetoviscosity to -19 cP at $H \sim 2000$ Oe and $\Omega \geq 700$ Hz. Later, Zeuner *et al.*,[20] conducted experiments similar to these of Bacri *et al.*, and compared their experimental results of reduced viscosity for a colloidal suspension of magnetite (Fe_3O_4) with the predictions of a theoretical model given in Ref. 19 for finite values of α. This model was based on the ferrohydrodynamics equations and the effective field method. They observed, for a large range of magnetic field strength and oscillation frequencies, the negative viscosity effect and found qualitative agreement between measurements and theory. The dependence of η^m on the flow vorticity in an oscillating magnetic field was studied experimentally by Gazeau *et al.*[21] for a ferrofluid in the Couette flow and later was demonstrated theoretically by Krekov *et al.*,[17] who solved numerically a set of ferrohydrodynamic equations showing the non-Newtonian behavior of a ferrofluid under conditions taking place in the experiments performed by Bacri *et al.*[19] On the other hand, Morimoto *et al.*[22] studied rheological and magnetic properties of a magnetic fluid composed of permanently magnetized spherical particles, subjected to both alternating magnetic and shear flow fields, by nonequilibrium Brownian dynamics simulations. The system was restricted to two dimensions and they took into account interparticle potentials, such as dipole-dipole interactions, van der Waals attraction, and repulsion caused by surfactant contact. Their results show the effect on rotational viscosity of the magnetic field strength and shear rate $\dot{\gamma}$ for Pe $\equiv \frac{3}{4}\tau_B \dot{\gamma} \geq 1$, exhibiting the negative viscosity effect. This decrease in viscosity was attributed to resonance between the rotational motions of the particles and the fluctuation of the magnetic field.

The response of magnetic fluids to rotating magnetic fields was first observed experimentally by Moskowitz and Rosensweig[23] and studied theoretically by Zaitsev and Shliomis.[24] Interest here has centered on measurements of velocity profiles of concentrated suspensions since field rotation causes macroscopic motion of the fluid due to the average rotation of the suspended particles, a phenomenon known as spin-up flow.[25] On the other hand, the rheological properties of magnetic fluids subjected to rotating magnetic fields can be studied experimentally from measurements of the torque required to rotate a spindle submerged in ferrofluid. Thus, Rinaldi *et al.*[26] showed for a ferrofluid composed of magnetite particles that with the magnetic field counter-rotating and corotating with the spindle, the torque increases and decreases, respectively, as the strength of the magnetic field and frequency increase (see Fig. 1). A decrease in the measured torque corresponds to a negative magnetoviscosity. In the experiments reported by Rinaldi *et al.* even negative torques can be measured, which would correspond to a negative effective viscosity of the suspension (that is, the magnetoviscosity is negative and larger in magnitude than the base fluid and Einstein viscosities). While there has been much work on the viscosity of magnetic fluids under oscillating magnetic fields, relatively few studies of the magnetoviscosity under rotating magnetic fields have been reported. Additionally, analytical models for magnetoviscosity in oscillating magnetic fields are based on the solution of the system of ferrohydrodynamic equations,[18,19,27] which are limited by the assumptions of the magnetization relaxation equations used.

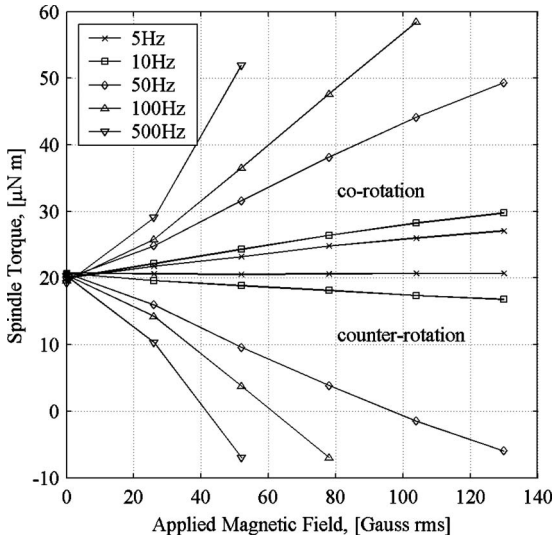


FIG. 1. Torque required to rotate a spindle surrounded with ferrofluid at a rate of 100 rpm as a function of applied field magnitude, rotation direction, and frequency. Reproduced with permission from C. Rinaldi *et al.*, J. Magn. Magn. Mater. **289**, 307 (2005). Copyright © 2005 Elsevier.

In this paper we present rotational Brownian dynamics simulations for the intrinsic magnetoviscosity $[\eta^m]$ of a magnetic fluid composed of noninteracting permanently magnetized spherical particles suspended in a Newtonian fluid and subjected to both magnetic fields and simple shear flow. The algorithm solves the noninertial particle angular momentum balance under an applied uniform oscillating/rotating magnetic field (i.e., there are no magnetic forces acting on the particles but there are external couples acting on them). We studied the effect of the amplitude and frequency of the external magnetic field and the magnitude of the shear rate on the intrinsic magnetoviscosity of the suspension. In Sec. II, we describe the particle model, present the equation of motion, describe our numerical model, and present the dimensionless form of the intrinsic magnetoviscosity. In Sec. III, the simulation parameters are summarized. In Sec. IV, we discuss the results obtained from simulations. In Sec. V we include concluding remarks.

II. ROTATIONAL BROWNIAN MOTION OF A PARTICLE IN MAGNETIC AND SHEAR FIELDS

For a suspended particle physical and magnetic properties are most naturally written in a Cartesian coordinate system with axes aligned with the principal semiaxes of the particle (primed axes) rather than relative to the coordinate axes of the laboratory space (unprimed axes), as illustrated in Fig. 2. We consider that the magnetic dipole moment of the particle m' is directed along the z'-axis, the magnetic field $\mathbf{H}$ is applied on the yz-plane, and the simple shear flow is along the y-axis. If the shear rate is denoted by $\dot{\gamma}$, then the

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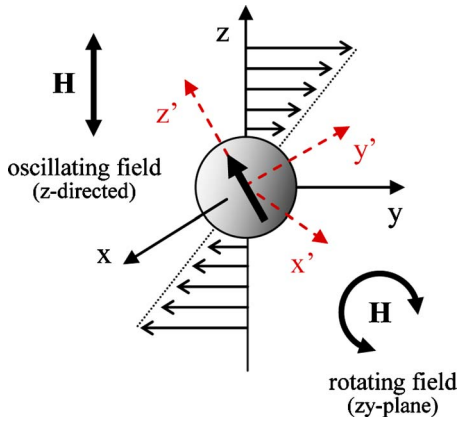


FIG. 2. (Color online) Particle model and coordinate axes.

unperturbed flow velocity $\mathbf{v}$ and the angular velocity of the fluid $\boldsymbol{\omega}_f$ are given by

$$\mathbf{v} = \dot{\gamma} z \mathbf{i}_y, \quad \boldsymbol{\omega}_f = \tfrac{1}{2} \nabla \times \mathbf{v} = -\tfrac{1}{2} \dot{\gamma} \mathbf{i}_x. \tag{2}$$

A. Rotational Brownian dynamics

Because of the small moment of inertia of the particles in a colloidal suspension,[28] we performed Brownian dynamics simulations based on the integration of the inertialess limit of the stochastic angular momentum equation

$$\mathbf{T}'_h + \mathbf{T}'_m + \mathbf{T}'_B = \mathbf{0}, \tag{3}$$

where $\mathbf{T}'_h = -\xi_r(\boldsymbol{\omega}' - \boldsymbol{\omega}'_f)$ is the hydrodynamic torque, $\mathbf{T}'_m = \mu_0(\mathbf{m}' \times \mathbf{H}')$ is the magnetic torque, and $\mathbf{T}'_B$ is the stochastic Brownian torque. Here the prime indicates a vector expressed relative to body-fixed axes. $\mathbf{H}' = \mathbf{A} \cdot \mathbf{H}$ is the magnetic field transformed to the body-fixed axes through the transformation matrix $\mathbf{A}$. In our case, orientation is parametrized using the Euler parameters e_0, e_1, e_2, and e_3,[14,29]

$$\mathbf{A} = \begin{bmatrix} e_0^2 + e_1^2 - e_2^2 - e_3^2 & 2(e_1 e_2 + e_0 e_3) & 2(e_1 e_3 - e_0 e_2) \\ 2(e_1 e_2 - e_0 e_3) & e_0^2 - e_1^2 + e_2^2 - e_3^2 & 2(e_2 e_3 + e_0 e_1) \\ 2(e_1 e_3 + e_0 e_2) & 2(e_2 e_3 - e_0 e_1) & e_0^2 - e_1^2 - e_2^2 + e_3^2 \end{bmatrix}. \tag{4}$$

Assuming that the Stokes law applies, Eq. (3) yields

$$-\xi_r(\boldsymbol{\omega}' - \boldsymbol{\omega}'_f) + \mu_0(\boldsymbol{\mu}' \times \mathbf{H}') + \mathbf{T}_B' = \mathbf{0}, \tag{5}$$

where $\xi_r = 8\pi\eta_0 r^3$ is the rotational hydrodynamic resistance, with η_0 as the viscosity of the carrier fluid and r as the radius of the particle, $\boldsymbol{\omega}'$ is the angular velocity of the particle, and μ_0 is the permeability of free space.

Solving Eq. (5) for $\boldsymbol{\omega}'$, setting $d\tilde{\boldsymbol{\Phi}}' = \tilde{\boldsymbol{\omega}}' d\tilde{t}$, where $d\tilde{\boldsymbol{\Phi}}'$ is the infinitesimal rotation vector, integrating in time $\tilde{t}$ using a first-order forward Euler method, and applying the fluctuation-dissipation theorem[30] to the Brownian term, we obtain

$$\Delta\tilde{\boldsymbol{\Phi}}' = \mathrm{Pe}_r \tilde{\boldsymbol{\omega}}'_f \Delta\tilde{t} + \alpha(\tilde{\mathbf{m}}' \times \tilde{\mathbf{H}}')\Delta\tilde{t} + \sqrt{2}\tilde{\mathbf{w}}', \tag{6}$$

where $\mathrm{Pe}_r = \dot{\gamma}/D_r$ is the rotational Peclet number with $D_r = k_B T/\xi_r$ as the rotational diffusion coefficient and α is the Langevin parameter. Vector $\tilde{\mathbf{w}}$ is a random vector characterized by a Gaussian distribution with zero mean $\langle\tilde{\mathbf{w}}\rangle = 0$ and variance $\langle\tilde{\mathbf{w}} \cdot \tilde{\mathbf{w}}\rangle = \Delta\tilde{t}$.

In the derivation of Eq. (6), the following scaled variables were used:

$$\tilde{\boldsymbol{\omega}}' = \frac{\boldsymbol{\omega}'}{D_r}, \quad \tilde{\boldsymbol{\omega}}'_f = \frac{\boldsymbol{\omega}'_f}{\dot{\gamma}}, \quad \tilde{\mathbf{m}}' = \frac{\mathbf{m}'}{m}, \quad \tilde{\mathbf{H}}' = \frac{\mathbf{H}'}{H}, \quad \tilde{\Omega} = \frac{\Omega}{D_r}, \tag{7}$$

where m and H are the magnitudes of m' and $\mathbf{H}'$, respectively, and $\tilde{\Omega}$ is the dimensionless magnetic field frequency. The particle angular velocity is nondimensionalized with respect to D_r as time is nondimensionalized with respect to the rotational diffusion time scale. The fluid vorticity is nondimensionalized with respect to $\dot{\gamma}$ in order to introduce the Peclet number in Eq. (6).

B. Calculation of the magnetoviscosity of the suspension

For polar fluids, such as ferrofluids, when the particles are acted upon by external couples, the resulting suspension-scale viscous (or deviatoric) stress tensor is characterized by both symmetric and antisymmetric parts.[31,32] The antisymmetric component $\boldsymbol{\tau}^a$ due to the action of magnetic couples arises from the slip between the local angular velocity of the suspension and the average angular velocity of the particles.[31] This can be obtained from[10,13]

$$\boldsymbol{\tau}^a = -\frac{n}{2}\langle \boldsymbol{\varepsilon} \cdot \mathbf{T}_m \rangle, \tag{8}$$

in which n is the number density of particles, $\boldsymbol{\varepsilon}$ is the alternating unit tensor, and $\mathbf{T}_m$ is the magnetic torque referred to space fixed axes. The brackets on the right hand side of Eq. (8) denote an average over the ensemble of particles in the suspension.

We are interested in the influence of the shear rate and the strength and frequency of the magnetic field on the magnetoviscosity of the suspension. For the simple shear flow given in Eq. (2) (see also Ref. 33), the magnetoviscosity of the suspension is given by $\eta_{zy}^m = \tau_{zy}^a/\dot{\gamma}$, and the intrinsic magnetoviscosity $[\eta_{zy}^m]$ is defined as

$$[\eta_{zy}^m] = \lim_{\phi \to 0} \frac{\eta_{zy}^m}{\phi \eta_0}. \tag{9}$$

Combining Eqs. (8) and (9) yields in dimensionless form

$$[\eta_{zy}^m] = -3\frac{\alpha}{\mathrm{Pe}_r}\langle \boldsymbol{\varepsilon} \cdot (\tilde{\mathbf{m}} \times \tilde{\mathbf{H}}) \rangle_{zy}. \tag{10}$$

Equation (10) furnishes $[\eta_{zy}^m]$ as a function of the Langevin parameter, the dimensionless shear rate expressed as a rotational Peclet number, and the average orientation of the particles.

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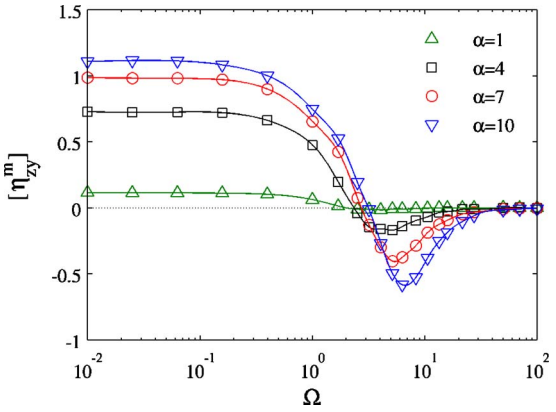


FIG. 3. (Color online) Dependence of the intrinsic magnetoviscosity on the magnitude and frequency of an oscillating magnetic field for $Pe_r=0.1$.

III. SIMULATION PARAMETERS AND CONDITIONS

Three types of simulation runs were performed: (i) response to an oscillating magnetic field $\tilde{\mathbf{H}}=\cos(\tilde{\Omega}\tilde{t})\mathbf{i}_z$, (ii) response to a corotating magnetic field $\tilde{\mathbf{H}}=\sin(\tilde{\Omega}\tilde{t})\mathbf{i}_y+\cos(\tilde{\Omega}\tilde{t})\mathbf{i}_z$, and (iii) response to a counter-rotating magnetic field $\tilde{\mathbf{H}}=\sin(\tilde{\Omega}\tilde{t})\mathbf{i}_y-\cos(\tilde{\Omega}\tilde{t})\mathbf{i}_z$. Here corotation means rotation in the same direction as the local vorticity of the fluid and counter-rotation means rotation in the opposite direction. All runs were performed starting from a random configuration, using 10^5 noninteracting particles, a time step of $\Delta\tilde{t}=0.01$, dimensionless frequencies $\tilde{\Omega}$ in the range of $10^{-2}-10^2$, and Langevin parameters of $1\leq\alpha\leq10$. For a particle of typical size $d\sim10$ nm suspended in a fluid with viscosity $\eta_0\sim10^{-3}$ Pa s, the Brownian relaxation time is very short $\tau_B\sim10^{-7}-10^{-6}$ s; therefore, large values of the Peclet number are difficult to obtain in practice, so Pe_r was varied from 0.1 to 1.

IV. RESULTS AND DISCUSSION

The effect of the strength and frequency of an oscillating magnetic field on the intrinsic magnetoviscosity $[\eta^m_{zy}]$ is shown in Fig. 3 for $Pe_r=0.1$. At low frequency of the magnetic field the intrinsic magnetoviscosity approaches an equilibrium value, see Ref. 14, and it decreases as the frequency increases. The intrinsic magnetoviscosity becomes negative at intermediate frequencies and tends to zero at high frequencies, where the magnetic field has no effect on the magnetoviscosity. As shown numerically by Morimoto *et al.*,[22] at low magnetic field frequencies the angular velocity of the particles is lower than the angular velocity of the fluid, which results in an increase in the apparent viscosity. On the other hand, at higher frequencies the angular velocity of the particles is higher than the angular velocity of the fluid, which results in a decrease of the apparent viscosity, as mentioned in Sec. I. At very high frequencies the particles are unable to follow the magnetic field fluctuation and therefore the particles rotate freely with the local vorticity of the carrier fluid. Furthermore, Fig. 3 shows that the minimum intrinsic magnetoviscosity decreases and the frequency corresponding to that minimum viscosity increases as the strength of the magnetic field, parametrized by α, increases.

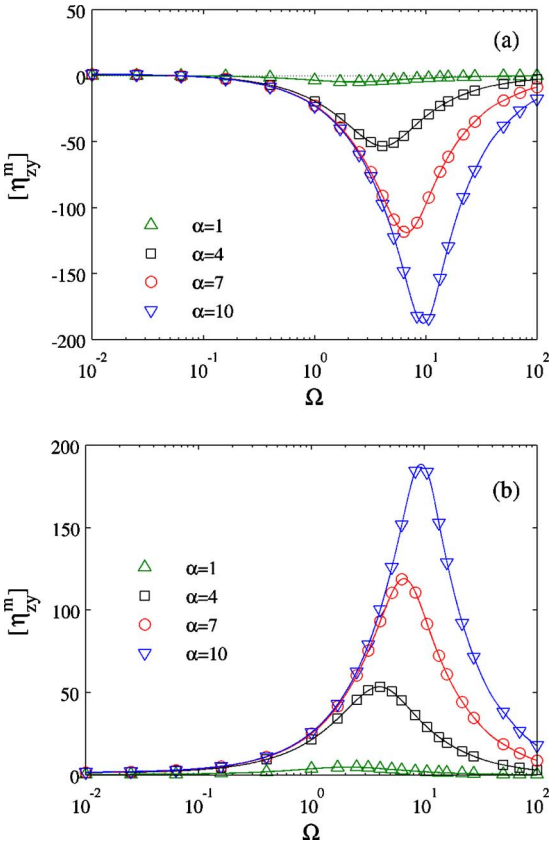


FIG. 4. (Color online) Dependence of the intrinsic magnetoviscosity on the magnitude and frequency of (a) a corotating and (b) a counter-rotating magnetic field for $Pe_r=0.1$.

Rotating magnetic fields exhibit a dramatic effect on the apparent viscosity of the suspension. Figure 4 shows the effect of the amplitude and frequency of a rotating magnetic field on the intrinsic magnetoviscosity for $Pe_r=0.1$. Results in Fig. 4(a) show that when a corotating magnetic field is applied the frequency of the magnetic field at which the intrinsic magnetoviscosity becomes zero is lower in comparison with that of the alternating field case. Furthermore, the minimum value of the viscosity is much lower and its magnitude dramatically increases as the strength α of the magnetic field increases. The field frequency required to reach the minimum intrinsic magnetoviscosity increases slightly as α increases. Also, the frequency range for under which negative intrinsic magnetoviscosity appears is wider than for oscillating magnetic fields. On the other hand, for a counter-rotating magnetic field, Fig. 4(b) shows that the magnetoviscosity is always positive in the whole range of frequencies, exhibiting a peak of maximum magnetoviscosity which increases in magnitude as α increases. As for corotating fields, the field frequency required to reach the maximum viscosity increases slightly as α increases. It is important to note that the curve of intrinsic magnetoviscosity for counter-rotating field is almost a mirror image of that for corotating fields, which could be attributed to the fact that the particles

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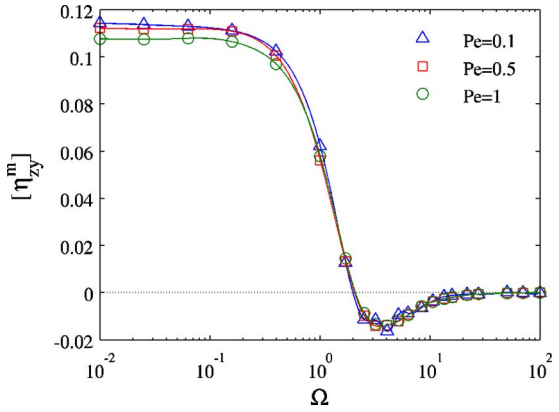


FIG. 5. (Color online) Dependence of the intrinsic magnetoviscosity on the Peclet number and frequency of an oscillating magnetic field for $\alpha=1$.

are rotating in opposite direction and to the reversibility of the low Reynolds number flow[33] imposed on the particles.

In contrast to the results obtained by Morimoto *et al.*[22] who considered $\mathrm{Pe}_r=1$ and 5, the Peclet number does not have a significant effect on the magnetoviscosity of the suspension in an oscillating magnetic field for $0.1\leq \mathrm{Pe}_r\leq 1$, as shown in Fig. 5. On the other hand, under corotating magnetic fields, Fig. 6(a) shows that for $\alpha=1$, the frequency where the intrinsic magnetoviscosity becomes zero is lower and the frequency range for negative viscosity is wider as Pe_r decreases. Furthermore, the field frequency required to reach

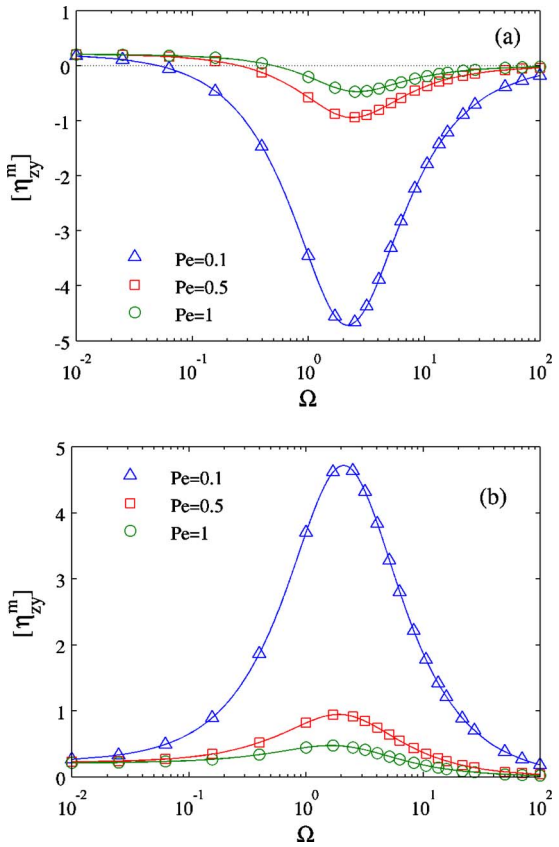


FIG. 6. (Color online) Dependence of the intrinsic magnetoviscosity on the Peclet number and frequency of (a) a corotating and (b) a counter-rotating magnetic field for $\alpha=1$.

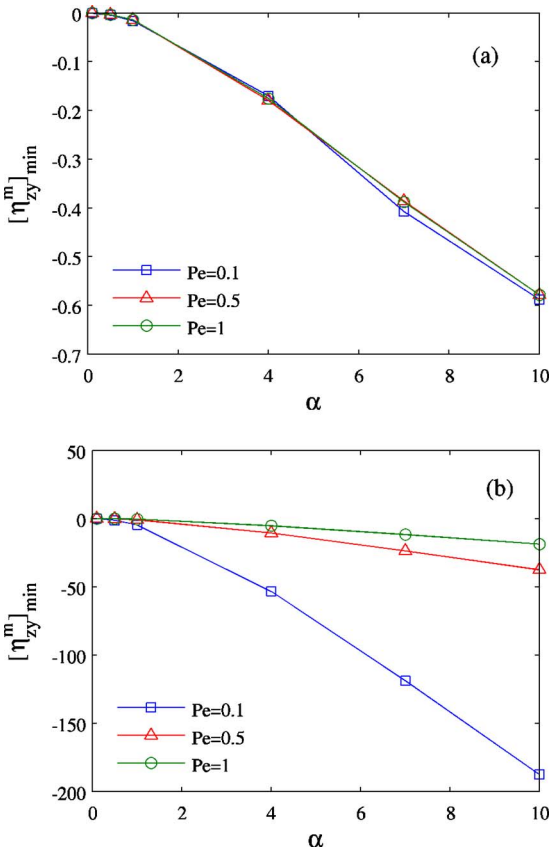


FIG. 7. (Color online) Minimum value of the intrinsic magnetoviscosity as a function of the Langevin parameter and Peclet number for (a) an oscillating and (b) a corotating magnetic field.

the minimum intrinsic magnetoviscosity remains almost constant as Pe_r increases. The effect of the shear rate on the intrinsic magnetoviscosity in counter-rotating field shows behavior similar to that with a corotating magnetic field [see Fig. 6(b)].

Figure 7 shows a comparison between the minimum values of the intrinsic magnetoviscosity obtained for both oscillating and corotating magnetic fields as a function of α and Pe_r. For both cases the figure shows that the magnitude of the intrinsic magnetoviscosity increases as the strength of the magnetic field increases. In the simulated range the rotational Peclet number has no effect on $[\eta^m_{zy}]_{\min}$ for an oscillating magnetic field, Fig. 7(a), whereas in a corotating field, Fig. 7(b) shows that the magnitude of $[\eta^m_{zy}]_{\min}$ increases as the Peclet number decreases.

It is important to note that for $\mathrm{Pe}_r=0.1$ the magnitude of the minimum intrinsic magnetoviscosity for a corotating magnetic field is $\sim 10^2$ times greater than that for an oscillating magnetic field. Thus, it may be possible to obtain negative values of the effective viscosity η_{eff} of a suspension composed of permanently magnetized spherical particles. The effective viscosity of the suspension, subjected to an external magnetic field, would be written as

$$\frac{\eta_{\mathrm{eff}}}{\eta_0}=1+\phi\left(\frac{5}{2}+[\eta^m_{zy}]\right). \tag{11}$$

Equation (11) is plotted in Fig. 8 as a function of the

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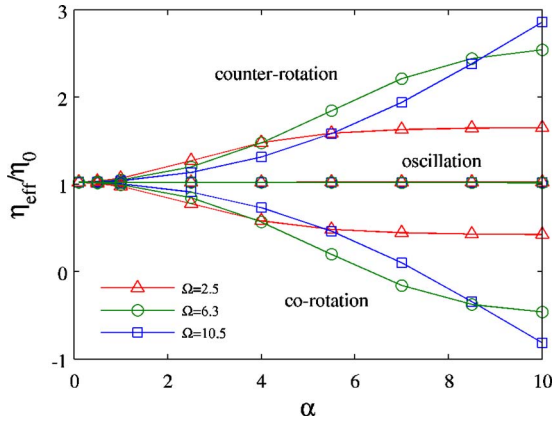


FIG. 8. (Color online) Normalized effective viscosity of the suspension as a function of the Langevin parameter for $Pe_r=0.1$, $\phi=0.01$, and various values of field frequency $\tilde{\Omega}$.

Langevin parameter for a particle volume fraction of $\phi=0.01$, which we assume as a dilute enough suspension. Figure 8 shows $\eta_{eff}<0$ for high values of α and frequency under corotating magnetic field. This result is analogous to the torque measurements on a commercial ferrofluid in rotating magnetic fields carried out by Rinaldi *et al.*[26] This effect has been described as if the fluid is filled with nanosized rotors that reduce the friction between adjacent fluid layers[21] due to the transfer of part of the magnetic field energy as kinetic energy into the fluid, leading to a decrease in the effective viscosity of the suspension. Figure 8 also shows a negligible effect of the oscillating magnetic field on η_{eff} in comparison with the case of rotating magnetic fields.

In the negative intrinsic magnetoviscosity regions the mean angular velocity of the particles should be higher than the local angular velocity of the carrier fluid. To illustrate this, we determined the dimensionless mean angular velocity of the particles $\langle\tilde{\omega}_x\rangle$ and compared it with the local angular velocity of the carrier fluid $\tilde{\omega}_f=1/2$. The time variation in the mean angular displacement $\langle\tilde{\Phi}_x(\tilde{t})\rangle$ about the x-axis is shown in Fig. 9 for $\alpha=4$ and $Pe_r=1$. Note that the slope in Fig. 9 gives the dimensionless mean angular velocity of the particles. In the case of a corotating magnetic field, Fig. 9(a) shows for $\tilde{\Omega}=0.1$ that the mean angular velocity of the particles is lower than the fluid angular velocity, $|\langle\tilde{\omega}_x\rangle|<\tilde{\omega}_f$, which leads to positive values of the intrinsic magnetoviscosity. As the field frequency increases, the mean angular velocity of the particles eventually is higher than the angular velocity of the fluid, $|\langle\tilde{\omega}_x\rangle|>\tilde{\omega}_f$, and the magnetoviscosity becomes negative. At very high magnetic field frequencies the particles cannot follow the magnetic field rotation, therefore they rotate freely with the fluid, $|\langle\tilde{\omega}_x\rangle|=\tilde{\omega}_f$, and the effect of the rotating magnetic field on the intrinsic magnetoviscosity disappears. In the case of a counter-rotating magnetic field, Fig. 9(b) shows that for $\tilde{\Omega}=0.1$ the particles rotate slower than in the case of a corotating field. This is because the hydrodynamic torque dominates the magnetic torque. As the frequency increases, the magnetic torque dominates and the particles rotate in opposite direction to the local vorticity, therefore the intrinsic magnetoviscosity increases. At very high frequencies the magnetic field has no effect on $[\eta_{zy}^m]$.

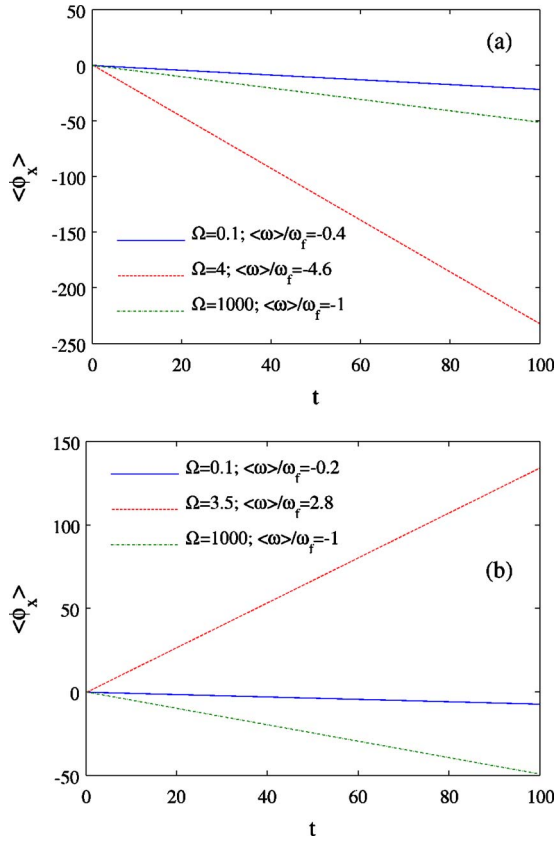


FIG. 9. (Color online) Time variation in the mean angular displacement of magnetic particles in both (a) corotating and (b) counter-rotating magnetic fields for $\alpha=4$ and $Pe_r=1$, and various values of field frequency $\tilde{\Omega}$.

The results shown in Fig. 9(a) qualitatively agree with those obtained by Morimoto *et al.*[22] for oscillating magnetic fields. We compared the dimensionless mean angular velocity of the particles for $\alpha=1$ and $Pe_r=1$, when they are subjected to oscillating and corotating magnetic fields. Figure 10 shows that for $\tilde{\Omega}=2.5$, in an oscillating magnetic field, the particles rotate at the same angular velocity of the fluid, but

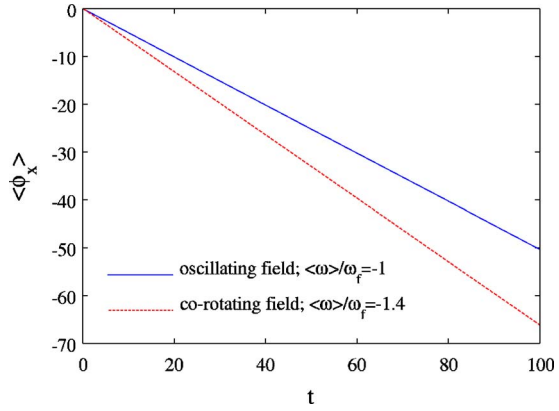


FIG. 10. (Color online) Comparison of variations of the mean angular displacement of magnetic particles in oscillating and corotating magnetic fields at $\tilde{\Omega}=2.5$ for $\alpha=1$ and $Pe_r=1$.

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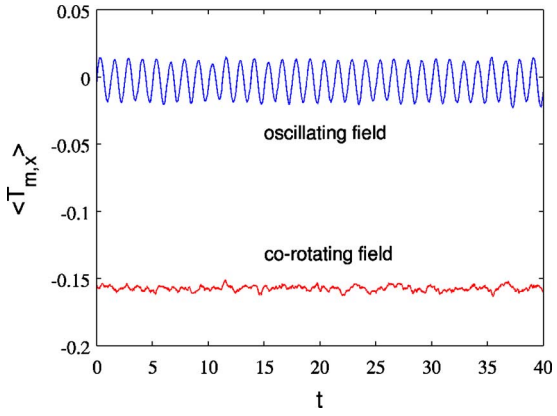


FIG. 11. (Color online) Variation in the macroscopic nondimensional magnetic torque ($\langle\tilde{\mathbf{T}}\rangle=\tilde{\mathbf{m}}\times\tilde{\mathbf{H}}$, where $\tilde{\mathbf{m}}$ is the dimensionless magnetization of the suspension) in the flow vorticity direction (x-axis) with time in an oscillating and a corotating magnetic field at $\tilde{\Omega}=2.5$ for $\alpha=1$ and $\mathrm{Pe}_r=1$.

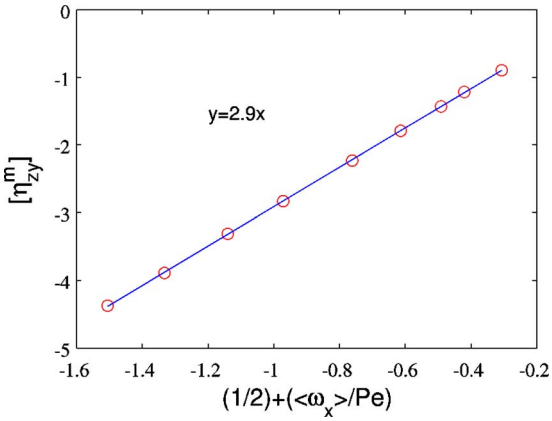


FIG. 12. (Color online) Intrinsic magnetoviscosity in corotating magnetic field as function of the angular slip velocity for $\alpha=1$ and $\mathrm{Pe}_r=0.1$.

in a corotating magnetic field the particles rotate faster than the fluid, which leads to lower values in the negative intrinsic magnetoviscosity. The reason is that as shown in Fig. 11, for an oscillating magnetic field the magnetic torque changes direction as the particles rotate and the field oscillates, at times opposing the rotation of the particles; whereas for a corotating magnetic field the magnetic torque always reinforces rotation in the direction of the vorticity, increasing the rotation rate of the particles.

The results from these simulations can also be used to verify the suspension-scale constitutive equation governing the antisymmetric part of the deviatoric stress, which according to Condiff and Dahler[31] is given by

$$\boldsymbol{\tau}^a=\tfrac{1}{2}\eta_v\boldsymbol{\varepsilon}\cdot[(1/2)\,\boldsymbol{\nabla}\times\mathbf{v}-\langle\boldsymbol{\omega}\rangle]. \tag{12}$$

Here $\eta_v=6\eta_0\phi$ is the vortex viscosity[34] and $\langle\boldsymbol{\omega}\rangle$ is the average angular velocity of the particulate phase of the suspension. Combining Eqs. (2), (7), (9), and (12) we obtain, in dimensionless form, that

$$[\eta^m_{zy}]=3\left(\frac{1}{2}+\frac{\langle\tilde{\omega}_x\rangle}{\mathrm{Pe}}\right). \tag{13}$$

Figure 12 clearly demonstrates that the intrinsic magnetoviscosity obtained by Brownian dynamics simulations agrees well with the theoretical prediction for dilute suspensions given by Eq. (13); hence, this result confirms the correctness of the constitutive equation (12) for the antisymmetric stress. A similar result was obtained by Feng *et al.*[32] by numerical boundary element method simulations but neglecting particle Brownian motion.

V. CONCLUSIONS

Using rotational Brownian dynamics simulations we have studied the rheological properties, specifically the intrinsic magnetoviscosity, of dilute magnetic fluids composed of magnetized spherical particles in oscillating/rotating magnetic fields and simple shear flow. The governing equation of the infinitesimal rotation vector was derived from the stochastic angular momentum balance and solved numerically. An intrinsic viscosity due to the magnetic field was calculated. The simulations have been carried out for a wide range of conditions in order to compare the effect of alternating and rotating magnetic fields.

Results for oscillating magnetic fields are in agreement with those obtained in Ref. 22, showing additionally that the Peclet number does not have a significant effect on the magnetoviscosity for $\mathrm{Pe}_r\leq 1$. Simulation results show a considerable effect of a corotating magnetic field on the intrinsic magnetoviscosity in comparison with an oscillating magnetic field. In counter-rotating magnetic fields the intrinsic magnetoviscosity was positive in the whole range of field amplitude and frequency and exhibits a maximum at an intermediate frequency. In this case the particles rotate slower or in opposite direction to the local angular velocity of the fluid. For corotating fields the intrinsic magnetoviscosity becomes zero at lower frequency and the minimum viscosity obtained is much lower at the same values of α and Pe_r parameters than for an oscillating magnetic field. In the negative intrinsic magnetoviscosity regions, the mean angular velocity of the particles is higher than the local angular velocity of the fluid. In corotating magnetic fields particles rotate faster than in oscillating magnetic fields, which results in higher magnitude of the negative magnetoviscosity. Therefore, corotating magnetic fields seem to be suitable to obtain negative effective viscosity in magnetic nanoparticles suspensions. Finally, simulation results for the intrinsic magnetoviscosity show that the commonly accepted constitutive equation for the antisymmetric part of the viscous stress tensor is applicable to dilute suspensions of Brownian spherical particles.

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